LUMIVAARA LOGISTICS OY

Company Overview and Operations Description

Background material for the FAIR AI assessment

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Lumivaara Logistics Oy is a fictional company created for report-generation examples. All names, figures and systems are invented.

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1. Company profile

Lumivaara Logistics Oy is a Finnish family-owned third-party logistics (3PL) company headquartered in Vantaa. The company provides contract warehousing, domestic road freight, e-commerce fulfilment, customs clearance and value-added logistics services to business customers across Finland. Operations are run from three warehouses in Vantaa, Tampere and Oulu with a combined warehouse area of 33,500 m².

The company employs 134 people (full-time equivalents at the end of 2025), complemented by temporary agency staff during peak seasons. Revenue in 2025 was EUR 24.6 million. Lumivaara's value proposition is “reliable logistics with a personal touch”: customers have a named contact person, and the company is known for flexible handling of irregular volumes and special requests.

> **Mission:** We make our customers' supply chains work every day, so they can focus on their own business.
> **Vision 2028:** The most reliable and efficient mid-sized logistics partner in Finland, growing profitably through smart use of technology.

2. History and milestones

Year	Milestone
2009	Company founded by Pertti and Leena Lindholm in Vantaa with six trucks and a 4,000 m² warehouse.
2012	First contract-warehousing customer in the retail sector; warehouse expanded to 9,000 m².
2015	Tampere warehouse opened to serve industrial customers in Pirkanmaa.
2017	EDI connections with the first five key customers.
2018	Cloud transport management system (TMS) taken into use.
2019	New Vantaa main warehouse (18,000 m²) and head office completed. Document scanning started.
2020	Warehouse management system (WMS) upgraded to a new version with a new database structure.
2021	Generation change: Kaisa Lindholm appointed CEO. E-commerce fulfilment service launched.
2022	ERP changed to Microsoft Dynamics 365 Business Central.
2023	Oulu warehouse opened. RPA bot for purchase-invoice matching. Customs clearance service launched.
2024	GDP-certified temperature-controlled area (1,200 m²) for pharma and healthcare customers in Vantaa.
2026	Microsoft 365 Copilot pilot for office staff. Strategy update and start of AI assessment with FAIR.

3. Services

3.1 Contract warehousing

Storage and handling of customers' goods in ambient, chilled (+2…+8 °C) and GDP-controlled (+15…+25 °C) conditions. Services include receiving, quality checks, putaway, inventory management, picking and packing, and dispatch. Typical contracts run for three years with a monthly fee per pallet position and per-transaction handling fees.

3.2 Domestic road freight

Full-truckload (FTL) and less-than-truckload (LTL) transport throughout Finland, with daily distribution routes in the Helsinki metropolitan area, Pirkanmaa and Northern Ostrobothnia. Linehaul between the three warehouses runs every weekday night. About 55 % of transport volume is run with the company's own fleet and 45 % with subcontracted hauliers.

3.3 E-commerce fulfilment

Order fulfilment for online retailers: integration with web shops, same-day dispatch for orders received by 15:00, consumer returns handling, and parcel carrier hand-over. E-commerce volumes are highly seasonal; November and December typically account for 28 % of annual e-commerce order lines.

3.4 Customs clearance

Import and export declarations for customers importing from outside the EU, mainly from Asia and the United Kingdom. Declarations are prepared by two licensed customs specialists using the Finnish Customs electronic service.

3.5 Value-added services

Labelling, kitting and assembly of product sets, repacking, returns inspection and campaign display building for retail customers.

4. Facilities and fleet

Site	Opened	Area (m²)	Pallet positions	Temperature zones	Staff (FTE)
Vantaa (HQ)	2019	18,000	21,500	Ambient, chilled, GDP	79
Tampere	2015	9,500	11,200	Ambient	34
Oulu	2023	6,000	6,800	Ambient, chilled	21
Total		**33,500**	**39,500**		**134**

The company's own fleet consists of 38 trucks: 14 semi-trailer combinations for linehaul, 18 distribution trucks (six of them with temperature control) and 6 vans for urban deliveries. The average fleet age is 5.4 years. All trucks have telematics units reporting position, fuel consumption and driving behaviour. Around 25 subcontracted hauliers complement the fleet, most of them small Finnish and Baltic companies.

Warehouse equipment includes 46 forklifts and reach trucks, handheld barcode scanners for all warehouse staff, and a semi-automated packing line for e-commerce orders in Vantaa.

5. Customers and market segments

The company has approximately 210 active business customers. The 15 largest customers account for 58 % of revenue and are connected by EDI. Customer relationships are long-term: the average customer relationship has lasted 6.8 years.

Segment	Share of revenue 2025	Typical customers	Main services used
Retail and e-commerce	41 %	Clothing, home goods and electronics retailers; online shops	Warehousing, e-commerce fulfilment, distribution
Industrial spare parts	27 %	Machinery and equipment manufacturers, service companies	Warehousing, express deliveries, FTL
Chilled food and beverage	18 %	Food producers, beverage importers	Chilled warehousing, temperature-controlled distribution
Pharma and healthcare	9 %	Medical device distributors, pharmaceutical wholesalers	GDP warehousing, controlled transport
Other	5 %	Public sector, events, construction	Project logistics, storage

The pharma and healthcare segment is the fastest-growing (+31 % in 2025) and has the highest margin. Retail customers apply the strongest price pressure, and several large retail contracts are due for renewal in 2027.

6. Organisation and personnel

Function	Head	FTE	Notes
Management	Kaisa Lindholm, CEO	3	CEO, CFO, HR manager
Warehouse operations	Mikko Saarinen, Operations Director	78	Three site managers, shift supervisors, warehouse workers
Transport	Mikko Saarinen	31	Traffic planners, drivers, fleet manager
Customer service	Anna Virtanen, Team Lead	7	Shared mailbox and phone service
Document handling	Mikko Saarinen	6	Freight document clerks at all three sites
Finance and invoicing	CFO	5	Invoicing, accounts payable, controlling
Sales	CEO	2	Key account managers
Customs	CFO	2	Licensed customs specialists
IT	Jari Koskela, IT Manager	1	Systems, reporting, user support; network and security outsourced
Total		**134**	

Average employee age is 38 years. Staff turnover in warehouse roles was 19 % in 2025, which makes training and standardised work instructions important. The company uses 15–40 temporary agency workers per day during the November–December peak.

7. Key figures 2023–2025

Figure	2023	2024	2025
Revenue (EUR million)	21.5	23.0	24.6
Operating profit (EBIT, EUR million)	0.97	0.85	0.79
EBIT margin	4.5 %	3.7 %	3.2 %
Personnel costs (EUR million)	8.9	9.6	10.4
Overtime costs (EUR thousand)	298	352	410
Temporary agency staff costs (EUR thousand)	640	715	802
Employees (FTE, year end)	121	129	134
Outbound order lines (million)	3.1	3.4	3.7
Shipments transported (thousand)	214	229	241
Investments in IT and digitalisation (EUR thousand)	185	120	95

> The declining EBIT margin is driven mainly by rising personnel costs (wage settlements, overtime and agency staff) and by price pressure in the retail segment. The board has set a target to return the EBIT margin to at least 5 % by 2028.

8. Information systems landscape

System	Purpose	In use since	Hosting	Integrations / access
Warehouse management system (WMS)	Receiving, putaway, inventory, picking, shipping	2012 (current version 2020)	On-premise, SQL Server	EDI, TMS (shipment hand-over), read-only SQL access for IT
RouteLine TMS	Transport orders, route planning, POD status, freight costs	2018	Cloud (SaaS)	REST API (single records only), WMS, driver mobile app
Microsoft Dynamics 365 Business Central	Finance, invoicing, purchasing	2022	Cloud	Manual and file-based import from TMS/WMS; RPA bot
EDI operator service	Orders, ASNs, invoices with 15 key customers	2017	External service	WMS, Business Central
Driver mobile app	Delivery confirmation, POD photos, signatures	2020	Part of TMS	TMS
Telematics	Vehicle position, fuel, driving behaviour	2019	Vendor cloud	Web portal, CSV export
Document archive	Scanned freight documents (PDF)	2019	File server (Vantaa)	Folder structure per site and month; no index or search
Microsoft 365	Email, Teams, SharePoint, Copilot pilot	2016	Cloud	Shared customer service mailbox
Power BI	Management reporting (5 reports)	2022	Cloud	Excel exports from WMS, TMS and Business Central
UiPath RPA	Purchase invoice to purchase order matching	2023	Cloud orchestrator	Business Central

There is no central data warehouse. Customer master data is maintained separately in the WMS, TMS and Business Central, and there is no shared customer identifier between the systems. Reporting relies on manual Excel exports prepared by the IT Manager.

9. Core process descriptions

9.1 Inbound (receiving)

1. The customer sends an advance shipping notice (ASN) by EDI, email or the customer portal. About 35 % of inbound deliveries arrive without an ASN.
2. The truck arrives at the gate; the gate clerk checks the CMR waybill and assigns an unloading door.
3. Warehouse workers unload, count and inspect the goods, and record deviations (damaged pallets, shortages) on paper and in the WMS.
4. Goods are labelled and put away to locations suggested by the WMS.
5. The signed CMR is scanned at the end of the shift and filed in the document archive.

9.2 Outbound (picking and dispatch)

1. Customer orders arrive by EDI (62 % of order lines), web shop integration (27 %) or email/Excel (11 %). Email and Excel orders are keyed in manually by customer service.
2. The WMS releases picking waves; workers pick with handheld scanners.
3. Goods are packed, labelled and staged; the WMS creates the shipment and hands it over to the TMS.
4. Traffic planners assign shipments to routes or subcontractors in the TMS.
5. The driver confirms the delivery in the mobile app, collects a signature and photographs the POD if a paper document is used.

9.3 Freight document handling

Freight documents include CMR waybills, delivery notes, proof-of-delivery (POD) documents, packing lists and customs documents. Documents arrive by email (PDF), as photos from the driver app and subcontractors, and on paper at the warehouse gates. Document clerks open each document, find the matching shipment in the TMS and key in the delivery confirmation, package and weight data, pallet exchange information and any remarks. Invoicing-relevant data is also entered in Business Central.

The process is fully manual. Documents from subcontractors arrive with a delay of 2–10 days, and invoicing of a shipment cannot be completed until its POD has been processed. Remarks about damage or shortages written by hand on the documents are often noticed late, after the deadline for raising a claim against the haulier (7 days under the company's subcontracting terms).

9.4 Customer service

Customer service handles a shared mailbox and a phone line. Typical requests are shipment status enquiries, changes to bookings, delivery time requests, damage reports and claims, invoicing questions and quote requests. Agents look up shipment status manually in the TMS and the WMS.

There is no ticketing system; follow-up of open cases relies on email flags and personal notes. The target is to respond to all emails within four working hours.

9.5 Workforce planning

Shift supervisors plan staffing one week ahead using their experience, the previous weeks' volumes and the customer forecasts available. Only about a third of the key customers provide regular volume forecasts. Short-term gaps are covered with overtime or agency workers booked the previous afternoon at a premium price.

10. Key performance indicators

KPI	Target	2025 actual	Comment
Order picking accuracy	99.8 %	99.6 %	Errors concentrated in peak season
On-time delivery (own fleet)	97 %	95.1 %	Subcontractors 91.4 %
Inbound dock-to-stock time	< 24 h	27 h	Longer for deliveries without ASN
POD processed within 2 days of delivery	95 %	71 %	Subcontractor documents arrive late
Invoice correction rate	< 1 %	3.1 %	Mainly caused by document entry errors
Customer service email response time	4 h	6.2 h (avg), 9.4 h (Dec)	No ticketing system
Customer satisfaction (NPS)	+40	+34	Annual survey, 88 respondents
Warehouse staff turnover	< 12 %	19 %	

11. Strategy 2026–2028 and digitalisation goals

The board approved an updated strategy in June 2026. The strategy has four focus areas:

1. **Profitable growth** – grow revenue to EUR 30 million by 2028, especially in pharma/healthcare and industrial spare parts, while improving the EBIT margin to 5 %.
2. **Operational excellence** – reduce manual work in office processes by 30 % by the end of 2028 so that growth does not require additional office staff.
3. **Customer experience** – real-time visibility and faster answers for customers; NPS +45 by 2028.
4. **People** – reduce warehouse staff turnover and improve planning so that overtime is cut by a third.

11.1 Digitalisation and AI goals

- Map core processes and identify opportunities for automation and artificial intelligence (in progress, autumn 2026).
- Run at least one AI pilot in 2027 with a measurable business case.
- Introduce an AI usage policy and training for office staff.
- Improve data quality: shared customer master data and a common reporting database by 2028.
- Evaluate a ticketing system for customer service.

> **Budget:** The 2027 budget proposal earmarks **EUR 50,000** for digitalisation and AI pilots, in addition to ordinary IT operating costs (approx. EUR 210,000 per year). Further investment will be decided based on pilot results.

12. Data protection and information security

Lumivaara processes personal data of customer contact persons, its own employees and drivers, and consumers (recipients of e-commerce deliveries: names, addresses, phone numbers, sometimes delivery instructions). The company acts as a data processor for its e-commerce customers under data processing agreements. A privacy statement and a record of processing activities exist; the CFO acts as the data protection contact person. No data protection officer has been appointed.

Information security is managed together with an IT outsourcing partner: multi-factor authentication for Microsoft 365, endpoint protection, and backups of the on-premise WMS. The company has no written policy on the use of generative AI tools. Several customer contracts (especially in pharma) include clauses restricting the transfer of customer data outside the EU/EEA.

13. Known challenges and risks

Challenge / risk	Impact	Current mitigation
Manual document handling delays invoicing and causes errors	High	Additional clerk hired in 2024; weekend overtime at month end
Unpredictable volumes from key customers	High	Agency staff framework agreement
Dependence on one IT person	High	Partial documentation; outsourcing partner for infrastructure
Price pressure in retail contract renewals 2027	High	Cost-efficiency programme
Fragmented customer and master data	Medium	Manual reconciliation
Uncontrolled use of public AI tools with customer data	Medium	None yet – AI policy planned
Driver and warehouse staff availability	Medium	Recruitment co-operation with vocational schools

14. Appendix: Glossary

Term	Explanation
3PL	Third-party logistics: outsourced logistics services
ASN	Advance shipping notice sent before a delivery arrives
CMR	International consignment note for road transport
EDI	Electronic data interchange between companies' systems
FTL / LTL	Full truckload / less than truckload
GDP	Good Distribution Practice for medicinal products
POD	Proof of delivery: signed confirmation that goods were delivered
TMS / WMS	Transport management system / warehouse management system

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